



Scene

A Scene, in Computer Graphics, is a digital description of a virtual world to be Rendered.

Think of it as a recipe for an image: The Scene tells the Renderer what exists where, and the Renderer figures out how it looks.

Components

A Scene is made up of several fundamental components, often called Scene Elements or Entities:

- Geometry: 3D objects with shape and position
- Materials: Surface properties (color, texture, shininess, transparency)
- Lights: Light sources that illuminate the Scene
- Camera: The virtual eye through which the Scene is viewed
- Transformations: Position, rotation, scale of objects
- Hierarchy: How objects are grouped or related (parent/child)
- Environment: Background, fog, global effects

Rasterization vs Ray Tracing

The concept of Scene is identical in both methods, except how the Renderer uses it:

Rasterization

- The Scene is converted to triangles
- Objects are projected onto a 2D screen
- The GPU processes them in object order
- The Scene is often organized in a Scene Graph or Entity-Component System for efficiency

Ray Tracing

- The Scene is stored as a list of geometric primitives
- Rays are shot from the camera through each pixel
- The Ray Tracer checks intersections with all objects in the Scene along the ray's path
- This is more computationally expensive but more physically accurate